

Curriculum Vitae

Mahyar Karimi

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Contact Details

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Education

09.2023 – **M.Sc. in Computer Science**,
04.2026 Institute of Science and Technology, Klosterneuburg, Austria.
 Advisor: Prof. Thomas Henzinger
 Expected date of graduation: March 31, 2026

09.2018 – **B.Sc. in Computer Engineering**,
02.2023 University of Tehran, Tehran, Iran.
 Graduated as one of the top 10 students in my class.
 Thesis advisor: [Prof. Hossein Hojjat](#)

Notable Coursework (full mark is 20/20):

Software testing:	19.4,	Database design:	19.1,
Distributed systems:	17.8,	Cyber-physical systems:	16.3,
Software engineering:	17.8,	Internet engineering:	18.4,
Computer architecture:	20.0,	Operating systems:	19.5.

Research Experience

07.2022 – **Research intern**, Institute of Science and Technology Austria (ISTA)
09.2023 Worked under the supervision of Prof. Thomas Henzinger over two summers (2022 and 2023), supported in 2022 by an OeAD research scholarship.

In 2022, we designed monitors for quantitative fairness of online decision-making agents, modelling agent-environment interaction as a Markov chain. In 2023, we studied how neural networks can act as proofs, using PyTorch for network training and composition, and SMT solvers (Z3 and Marabou) for verification.

2021 – **Research assistant**, University of Tehran
2023 Collaborated with [Prof. Hossein Hojjat](#) and [Amir Hossein Seyhani](#).

We developed a model for causal reasoning over concurrent systems, with Winskel's *event structures* as our concurrency model, and the Halpern-Pearl definition of root cause. We have a Python tool for causality checking, with significant optimisations for verifying a candidate root cause.

Industry Experience

2022–3 **Software engineer**, [Divar](#).

I worked in the *customer trust* team, where we found ways to find and stop fraudulent users by tracing their behaviour across the platform. Before that, I worked in the *jobs* team, a new direction of the company for making a job market functionality.

Related skills: Microservices architecture, Django, CI/CD, Scrum, Kanban

2021 **Infrastructure intern**, [Tapsell](#).

Implemented a simple query caching mechanism, with Redis as the cache.

Related skills: Kubernetes, Redis, Nginx, Docker

Teaching Experience

2020 – University of Tehran

2022 Teaching assistant in the following courses:

- *Advanced Programming* (Lecturer: [Prof. Ramtin Khosravi](#))
For this course, the TAs would work in teams for designing assignments. Students would receive a review of their code and oral feedback from the TA during grading.
- *Data Structures* (Lecturer: [Prof. Fathiyeh Faghieh](#))

Extracurricular Coursework

05.2023 **Functional Program Design in Scala**

Lecturer: Prof. Martin Odersky (EPFL).

03.2023 **Functional Programming Principles in Scala**

Lecturer: Prof. Martin Odersky (EPFL).

11.2023 **Supervised Machine Learning: Regression and Classification**

Lecturer: Prof. Andrew Ng (course offered by DeepLearning.AI).

Presentations

09.2023 **Monitoring Markov Chains**

Presented at the 15th Alpine Verification Meeting, Prague, Czechia.

09.2024 **Monitors, Systems, and Privacy**

Presented at the 16th Alpine Verification Meeting, Freiburg, Germany.

Academic Service

2025 **Artifact evaluation committee member**, CAV 2025 (International Conference on Computer Aided Verification).

2025 **Subreviewer**, ECAI 2025 (European Conference on Artificial Intelligence) and FMCAD 2025 (Formal Methods in Computer-Aided Design).

Publications

Refereed Conference Papers

- [1] Thomas A. Henzinger, Mahyar Karimi, and K. S. Thejaswini. “Privacy-Preserving Runtime Verification”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. CCS ’25. Taipei, Taiwan: Association for Computing Machinery, 2025, pp. 2774–2787. ISBN: 9798400715259. DOI: [10.1145/3719027.3765137](https://doi.org/10.1145/3719027.3765137).
- [2] Thomas Henzinger et al. “Runtime Monitoring of Dynamic Fairness Properties”. In: *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. FAccT ’23. Chicago, IL, USA: Association for Computing Machinery, 2023, pp. 604–614. ISBN: 9798400701924. DOI: [10.1145/3593013.3594028](https://doi.org/10.1145/3593013.3594028).
- [3] Thomas A. Henzinger et al. “Monitoring Algorithmic Fairness”. In: *Computer Aided Verification*. Ed. by Constantin Enea and Akash Lal. Cham: Springer Nature Switzerland, 2023, pp. 358–382. ISBN: 978-3-031-37703-7.